

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Modeling Population Change

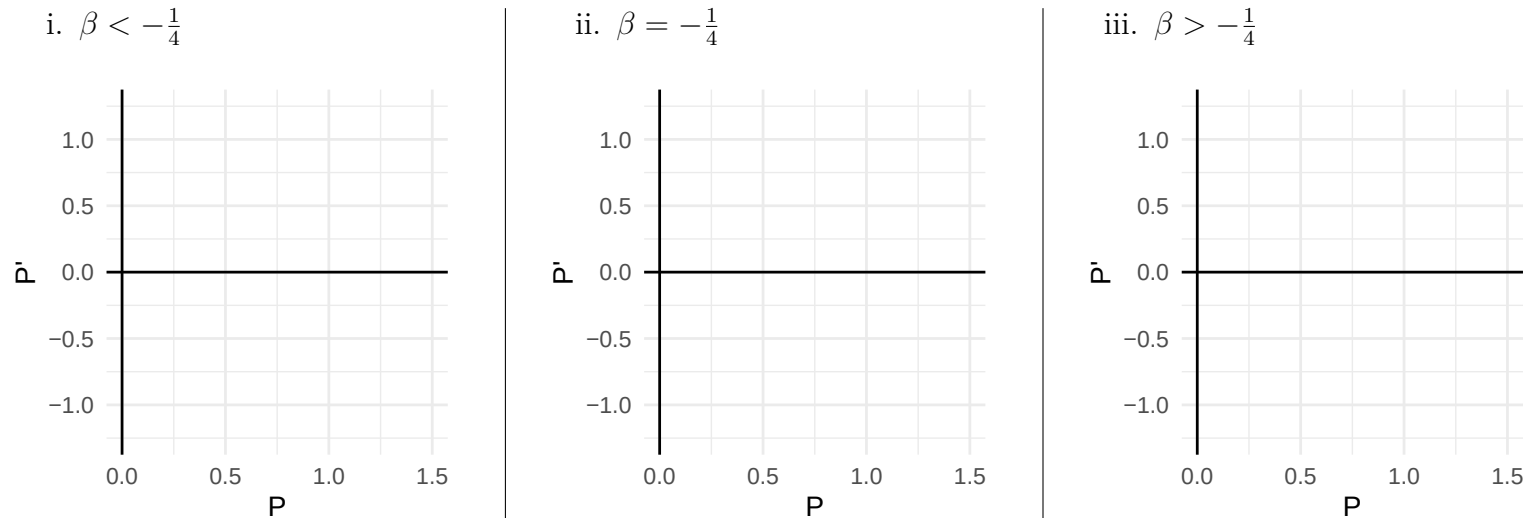
1. Population Dynamics

The population proportion P at time t , relative to the carrying capacity in an environment with limited resources, is modeled by the logistic growth/decay ODE,

$$P' = P(1 - P) + \beta,$$

where P' is the rate of change of the population, P is the natural rate of growth/decay, $(1 - P)$ represents the rate of decay due to environmental pressure, and β is the external growth/decay rate.

a. Use the blank axis below to draw complete and well-represented phase diagrams of the ODE as the parameter β changes for cases:



b. For each β case in Part (a), describe the general behavior of the population using the phase diagrams. Assume that the initial population starts at $0 < P(0) < 1$.

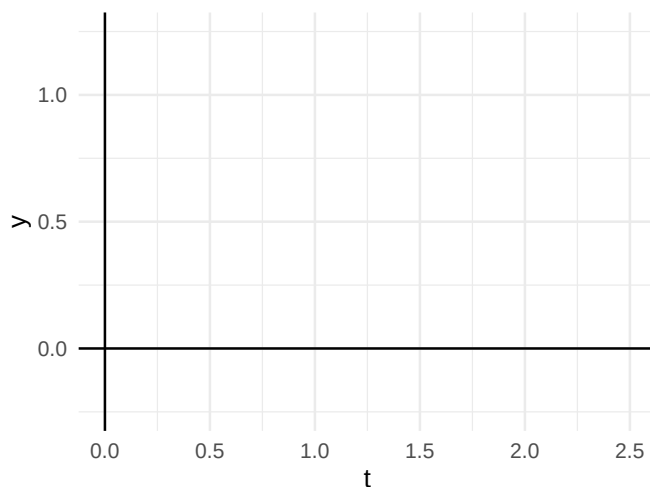
2. Logistic Growth Exact Solution

Continued from Problem (1). Consider the initial value problem,

$$P' = P(1 - P), \quad P(0) = \frac{1}{4}.$$

- a. Use the method of separation of variables to determine the general solution.

- c. Use the blank axis below to draw a complete and well represented slope field of the ODE. Then, draw the specific solution on the same field.



- b. Determine the specific solution.

- d. Determine the time it takes for the population to triple from its initial size.